

VENKATA ASISH MADDURI

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Professional Summary

Data Engineer with **6+ years of experience** building production data platforms, cloud-native backend services, and large-scale analytics pipelines. Expertise in Python, AWS, Apache Spark, FastAPI, computer vision, event-driven architectures, and scalable data engineering, with hands-on experience deploying LLM, RAG, and machine learning solutions. Proven track record of optimizing distributed systems, improving platform reliability, and delivering scalable data and AI solutions from development through production.

Technical Skills

- **Languages:** Python, SQL, PySpark
- **Generative AI:** LLMs, RAG, LangChain, LangGraph, OpenAI API, Gemini API, Prompt Engineering, Tool Calling, Structured Outputs
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, YOLOv11, OpenCV, Computer Vision, Model Deployment
- **Backend:** FastAPI, Flask, REST APIs, Microservices, Pydantic, Docker
- **AWS:** Lambda, S3, SQS, ECS, ECR, Batch, EMR, Glue, IAM, CloudWatch, Kinesis
- **Data Engineering:** Apache Spark, PySpark, Apache Iceberg, Kafka, Dagster, dbt, ETL/ELT, Data Modeling
- **Databases:** Snowflake, Amazon Redshift, PostgreSQL, Hive, Trino
- **DevOps:** Git, GitHub Actions, GitLab CI/CD, Terraform, CloudFormation, Linux

Professional Experience

Corteva Agriscience

Sept 2025 - Present
Des Moines / Johnston, IA

Data Engineer - Data Platform

- Built and maintained the production data platform supporting greenhouse computer vision workflows across wheat, corn, soybean, and sorghum experiments, processing more than **500 plant samples** and **50,000 hyperspectral measurements daily**.
- Deployed TensorFlow and YOLOv11 models into production for wheat growth-stage classification, wheat-head detection, and soybean-pod detection.
- Designed and implemented components of an event-driven processing architecture using Amazon S3, AWS Lambda, SQS, AWS Batch, Kafka, and Dagster for workload orchestration, model execution, and downstream processing.
- Optimized the production inference pipeline by identifying performance bottlenecks, reducing image-processing latency from approximately **1 minute to 20 milliseconds**.
- Led the migration of the production platform from Python 3.6 to Python 3.12, upgrading more than **50 dependencies** while maintaining compatibility across production models and data pipelines.
- Designed and implemented Apache Iceberg Bronze, Silver, and Gold data layers, optimizing PySpark transformations to reduce ETL runtime by **55%**, lower Amazon S3 storage costs by **40%**, and support more than **1 million records per day**.
- Developed automated validation and monitoring checks for production data pipelines, improving reliability by detecting processing failures, validating pipeline outputs, and reducing regression risk during platform upgrades.

Cognizant Technology Solutions

Jul 2019 - Jul 2023
Hyderabad, India

Data Engineer | Senior Data Engineer

- Built distributed PySpark pipelines on AWS EMR to process more than **500TB** of retail sales and transaction data across **2,500+ stores**, delivering scalable data pipelines for enterprise analytics and reporting.
- Developed a real-time inventory processing platform using Amazon Kinesis, AWS Lambda, and Spark Streaming to process more than **20 million events daily**, enabling near real-time inventory visibility and operational monitoring.
- Led the migration of more than **1PB** of Oracle and Teradata data to Snowflake and Amazon Redshift, redesigning data models, partitioning strategies, and transformation pipelines to improve query performance by **50%**.
- Designed reusable SQL models, dbt transformations, and automated data validation frameworks that improved data quality, simplified downstream reporting, and reduced manual reconciliation efforts.
- Optimized Spark jobs through partitioning, caching, and query tuning, reducing pipeline execution time and improving resource utilization across large-scale data processing workloads.
- Automated cloud infrastructure provisioning and CI/CD pipelines using Terraform, CloudFormation, and GitHub Actions, reducing deployment time by **55%** while improving release consistency across development, QA, and production environments.
- Collaborated with business stakeholders, analysts, and engineering teams to design scalable data architectures, optimize ETL workflows, and deliver reliable datasets supporting enterprise reporting and business intelligence initiatives.

AI Projects

CodeSentinel

- Built a guarded multi-agent software engineering platform using LangGraph, structured outputs, repository analysis, validation loops, and human approval.
- Implemented deterministic guardrails, dependency validation, unified diffs, backups, and safe file updates to prevent unintended AI-generated code changes.

EnterpriseRAG

- Built a document intelligence platform supporting PDF, Word, and text ingestion with FAISS semantic retrieval, FastAPI services, and source-grounded generation.
- Added retrieval evaluation, citation coverage, hallucination detection, latency and cost monitoring, and Docker-based deployment.

LLMOps Hub

- Developed an LLM evaluation platform supporting prompt versioning, experiment tracking, structured traces, and regression testing.
- Added observability for latency, token usage, inference cost, quality trends, failure categories, and production model monitoring.

Education

M.S. Data Science, University at Buffalo

May 2025

B.Tech. Electronics & Computer Engineering, Sreenidhi Institute of Science and Technology

Jun 2019

Certifications

AWS Certified Data Engineer - Associate